

Binomial Coefficients

unordered samples r out of n
w/o replacement in $\binom{n}{r}$ (nCr)

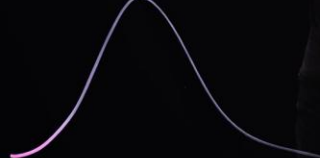
$$\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

$(x+y)^0$
 $(x+y)^1$
 $(x+y)^2$
 $(x+y)^3$
 $(x+y)^4$
 $(x+y)^5$



1 5 10 10 5 1



Multinomial Distribution

n obj. divided into r classes
of size $n_1, n_2, \dots, n_r$ (order not imp.)

not product

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!}$$

assumes $(n_1 + n_2 + \dots + n_r = n)$

Pf. $\binom{n}{n_1, n_2, \dots, n_r} = \binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \dots \binom{n-n_1-n_2-\dots-n_{r-1}}{n_r}$

$$= \frac{n!}{n_1!(n-n_1)!} \cdot \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \cdot \frac{(n-n_1-n_2)!}{n_3!(n-n_1-n_2-n_3)!} \dots \frac{(n-n_1-n_2-\dots-n_{r-1})!}{n_r!(0)!}$$

1. Pairing off $2n$ people:

If you have $2n$ people and want to know how many ways they can be formed into pairs, you are dividing $2n$ total objects into n groups, where each group (pair) has a size of 2. The formula becomes:

$$\frac{(2n)!}{2! \cdot 2! \dots 2!} = \frac{(2n)!}{(2!)^n}$$

This accounts for all the ways to partition the total group into these specific pairs.

2. Forming subcommittees from 7 people:

In this example, you have a total of $n = 7$ people and want to form three subcommittees of sizes $N_1 = 2$, $N_2 = 3$, and $N_3 = 2$ (since $2 + 3 + 2 = 7$). Plugging these into the multinomial formula:

$$\frac{7!}{2! \cdot 3! \cdot 2!} = \frac{5040}{2 \cdot 6 \cdot 2} = \frac{5040}{24} = 210$$

There are **210 distinct ways** to form these specific subcommittees.

For $(x+y)^3$, you calculate the coefficients using $n = 3$ for r values from 0 to 3. The general formula for each coefficient is $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ (1:25).

When expanding $(x+y)^3$, the coefficients are determined by calculating these combinations (3:23-5:30):

- **Term 1:** $\binom{3}{0} = 1$ (Coefficient for x^3)
- **Term 2:** $\binom{3}{1} = 3$ (Coefficient for x^2y)
- **Term 3:** $\binom{3}{2} = 3$ (Coefficient for xy^2)
- **Term 4:** $\binom{3}{3} = 1$ (Coefficient for y^3)

Adding these together, the full expansion is $1x^3 + 3x^2y + 3xy^2 + 1y^3$. As you noted, these values correspond directly to the rows in *Pascal's Triangle* (5:18).

Binomial Distribution: This is used when there are only **two possible outcomes** per trial (often called "success" or "failure"). For example, if you flip a coin 10 times, each flip can only be heads or tails. You are looking for the probability of getting exactly heads

Multinomial Distribution: This is a generalization used when there are **more than two possible outcomes** per trial. For instance, if you roll a six-sided die, each roll has six possible outcomes. If you want to know the probability of getting a specific count for each number over multiple rolls, you use the Multinomial distribution